

Building Simple Neuron Circuits

Javalab Connections of Neuron · Javalab

Year 7	~30 minutes	Science · Biology
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NOTICE — AI-generated worksheet

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Simulation: https://javalab.org/en/neuron_en/

Curriculum links: brain function · brain development

Learning intentions

- I can build a simple circuit of connected neuron gates (AND, OR, NOT).
- I can describe how a neuron's output depends on the signals it receives as input.
- I can explain, using this model, how connected neurons can combine information from more than one source.

Getting started

Open the simulation and find the AND, OR and NOT neuron gates. Start by wiring a single input into one gate and watch what output it produces, then build up to a small circuit with two or three connected gates for the later tasks.

Tasks

1. Wire two inputs into an AND gate. Turn each input on and off in different combinations and record when the gate fires (produces an output).

2. Repeat the previous task for an OR gate, recording the output for each combination of inputs.

Input 1	Input 2	Output

3. Describe the difference in behaviour between the AND gate and the OR gate.

4. What does the NOT gate do to a single input signal?

5. Connect two or three gates together to build a small circuit. Describe what your circuit does and under what input conditions it produces an output.

6. Real neurons receive signals from many other neurons at once and only 'fire' if enough input arrives together. Explain how the AND gate in this simulation is a simple model of that idea.

7. Name one way this simulation's model is a simplification of how real neurons actually work.

Extension (optional)

Design a circuit, using the gates in this simulation, that only produces an output when exactly one of two inputs is active, but not when both are active. Explain how you built it.
